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SEMICONDUCTOR DEVICE

Abstract

The present disclosure includes a substrate and a capacitor structure. The substrate includes a periphery region. The capacitor structure is disposed on the substrate and includes a plurality of capacitors arranged in an array, wherein at least two capacitors physically contact with each other along a direction being horizontal to the substrate. Through these arrangements, the semiconductor device enables to improve the possible structural defects of the semiconductor device caused by continuously increased cell-density, by arranging the dummy capacitors around the cell array, so as to obtain a more stable and more reliable structure to achieve better functions and performances.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present disclosure generally relates to a semiconductor device, and more particularly, to a semiconductor device including a capacitor structure.

2. Description of the Prior Art

[0002] With the trend of miniaturization of various electronic products, the design of semiconductor devices must also meet the requirements of high integration and high density. Under the current mainstream of development trend, dynamic random access memories (DRAMs) having recessed gate structures have gradually replaced the DRAMs having only planar gate structures due to longer carrier channel length for the same semiconductor substrate so as to reduce current leakage of capacitor structures. In general, a DRAM cell with a recessed gate structure includes a transistor component and a charge storage device to receive voltage signals from bit lines and word lines. However, due to the limitations of current processing technologies, there are still many defects in currently available DRAM cells with recessed gate structures, which need to be further improved to effectively improve the performance and reliability of related memory devices.

SUMMARY OF THE INVENTION

[0003] An object of the present disclosure is to provide a semiconductor device where a dummy capacitor is disposed around a storage cell array, such that, the possible structural defects of the semiconductor device caused by continuously increased cell-density will be improved. In this way, the semiconductor device may therefore obtain a more stable and more reliable structure, to achieve better functions and performances.

[0004] In order to achieve the above object, one embodiment of the present disclosure provides a semiconductor device including a substrate and a capacitor structure. The capacitor structure is disposed on the substrate and includes a plurality of capacitors arranged in an array, wherein at least two of the capacitors physically contact with each other in a horizontal direction horizontal to the substrate.

[0005] In order to achieve the above object, another embodiment of the present disclosure provides a semiconductor device including a semiconductor device including a substrate, a capacitor structure, and a supporting structure. The capacitor structure is disposed on the substrate and includes at least one first capacitor and at least one second capacitor. A bottommost surface of a bottom electrode layer of the at least one first capacitor is lower than a bottommost surface of a bottom electrode layer of the at least one second capacitor, and at least one of the at least one first capacitor and the at least one second capacitor includes a bending end. The supporting structure is disposed on the substrate and between the capacitors, the supporting structure includes a first supporting layer and a second supporting layer stacked sequentially from bottom to top.

[0006] Overall speaking, according to the semiconductor device of the present disclosure, dummy capacitors are arranged around the storage cell array, whose top electrode layers are in physical contact with each other, capacitor dielectric layer are in physical contact with each other, and/or top electric layers are in physical contact with each other, to improve the possible structural defects of the semiconductor device caused by continuously increased cell-density. Then, the semiconductor device is allowable to obtain a more reliable structure, to achieve better functions and performances.

[0007] These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The accompanying drawings are directed to provide a better understanding of the embodiments and are included as parts of the specification of the present disclosure. These drawings and descriptions are used to illustrate the principles of the embodiments. It should be noted that all drawings are schematic, and the relative dimensions and scales have been adjusted for the convenience of drawing. Identical or similar features in different embodiments are marked with identical symbols.

[0009] FIG. 1 to FIG. 2 are schematic diagrams illustrating a semiconductor device according to a first embodiment of the present disclosure, wherein:

[0010] FIG. 1 illustrates a top view of a semiconductor device; and

[0011] FIG. 2 illustrates a cross-sectional view of a semiconductor device.

[0012] FIG. 3 to FIG. 4 are schematic diagrams illustrating a semiconductor device according to a second embodiment of the present disclosure, wherein:

[0013] FIG. 3 illustrates a top view of a semiconductor device; and

[0014] FIG. 4 illustrates a cross-sectional view of a semiconductor device.

[0015] FIG. 5 is a schematic diagram illustrating a semiconductor device according to a third embodiment of the present disclosure.

[0016] FIG. 6 is a schematic diagram illustrating a semiconductor device according to a fourth embodiment of the present disclosure.

[0017] FIG. 7 is a schematic diagram illustrating a semiconductor device according to a fifth embodiment of the present disclosure.

[0018] FIG. 8 is a schematic diagram illustrating a semiconductor device according to a sixth embodiment of the present disclosure.

DETAILED DESCRIPTION

[0019] For better understanding of the presented disclosure, preferred embodiments will be described in detail. The preferred embodiments of the present disclosure are illustrated in the accompanying drawings with numbered elements. In addition, the technical features in different embodiments described in the following may be replaced, recombined, or mixed with one another to constitute another embodiment without departing from the spirit of the present disclosure.

[0020] Please refer to FIG. 1 to FIG. 2, which are schematic diagrams illustrating a semiconductor device **10** according to a first embodiment of the present disclosure, with FIG. 1 and FIG. 2 illustrating a top view and a cross-sectional view of the semiconductor device **10** respectively. Firstly, as shown in FIG. 1, the semiconductor device **10** includes a substrate **100** and a capacitor structure **150** disposed on the substrate **100**. The substrate **100** for example includes silicon substrate, silicon-containing substrate (for example including SiC or SiGe) or silicon-on-insulator (SOI) substrate. Also, the substrate **100** further includes a region with a relative higher elemental integration, for example being a cell region **104**, and another region with a relative lower elemental integration, for example being a peripheral region **102**. The cell region **104** and the periphery region **102** may be adjacent to each other, but not limited thereto. preferably, the peripheral region **102** is disposed around at the periphery of the cell region **104**, as shown in FIG. 1, but not limited thereto.

[0021] The capacitor structure **150** is disposed on the substrate **100**, and which includes a plurality of capacitors **140** alternately arranged in a horizontal direction D1 being horizontal to the substrate **100**, with the capacitors **140** being arranged along the horizontal direction D1 and a vertical direction D2 into an array. Capacitors **142** disposed within the cell region **104** are separately from each other by a supporting structure **160** disposed therebetween, so that, each capacitor **142** may be electrically connected to a transistor element (not shown in the drawings) within the substrate **100**,

via a plug (not shown in FIG. 1) underneath, thereby forming a storage cell array. It is noted that, at least two capacitors **144**, **146** disposed within the peripheral region **102** physically contact with each other in the horizontal direction **D1**, to function like a dummy capacitor. Thus, through arranging the dummy capacitors each physically contacting with another at the outer periphery of the storage cell array, the difference of the elemental density between the cell region **104** and the peripheral region **102** will be balanced while fabricating the semiconductor device **10**, thereby improving the possible structural defects of the semiconductor device caused by continuously increased cell-density. In this way, the semiconductor device **10** is allowable to obtain a more reliable structure, to achieve better functions and performances.

[0022] Precisely speaking, further in view of FIG. 1, the capacitor structure **150** includes a bottom electrode layer **152**, a capacitor dielectric layer **154**, and a top electrode layer **156** disposed in the horizontal direction **D1**, with the capacitor dielectric layer **154** being disposed at a side of the bottom electrode layer **152**, and with the top electrode layer **156** being further disposed at a side of the capacitor dielectric layer **154**, to configure as each capacitor **140**. In the present embodiment, the bottom electrode layer **152** for example includes an U-shaped bottom electrode (as shown in FIG. 2) having an U-shaped cross-section structure, and the top electrode layer **156** is filled in the gap between the U-shaped cross-section of the bottom electrode layer **152**. In one embodiment, the top electrode layer **156** may optionally include a monolayer structure or a multilayer structure, with the multilayer structure of the top electrode layer **156** being disposed at an inner side and an outer side of the bottom electrode layer **152**, as shown in FIG. 1. Accordingly, each capacitor **140** will present in a cylinder cross-sectional structure, but not limited thereto. In another embodiment, two capacitors **144** disposed within the peripheral region **102** physically contact with each other for example through the connecting bottom electrode layers **152**, the connecting capacitor dielectric layer **154**, or the connecting top electrode layer **156**, to function like the dummy capacitor as shown in FIG. 1, but not limited thereto. Alternately, at least one capacitor **146** disposed within the peripheral region **102** includes a relative smaller width **W1** in the horizontal direction **D1**, such that which may directly contact other capacitors **144** adjacent thereto. Through these arrangements, two or more than two capacitors **144**, **146** within the peripheral region **102** contacts with each other to function like the dummy capacitors, optionally through the contacting bottom electrode layers **152**, the contacting top electrode layers **156**, or the capacitor dielectric layers **154**.

[0023] On the other hands, as shown in the cross-sectional view of FIG. 2, the semiconductor device **10** includes a substrate **100**, the capacitor structure **150**, and a supporting structure **160** disposed on the substrate **100**, between each capacitor **140**. The supporting structure **160** precisely includes a first supporting layer **162** and a second supporting layer **164** stacked from bottom to top, for supporting the middle portion and the top portion of each capacitor **140** respectively, but not limited thereto. It is noted that, there is no plug disposed under the capacitors **144** with in the peripheral region **102**, so that, the capacitors **144** will not electrically connect to the transistor element in the substrate **100**, to function like the dummy capacitors thereby. The bottommost surfaces of the bottom electrode layers **152a**, **152b**, **152c**, of at least two capacitors **144a**, **144b**, **144c** to a top surface of the substrate **100** are in different heights in the vertical direction **D3**. Precisely, the bottommost surface of the bottom electrode layer **152a** of one capacitor **144a** is lower than the bottommost surface of the bottom electrode layer **152b** of another capacitor **144b**, with the bottom electrode layer **152a** of the capacitor **144a** and/or the bottom electrode layer **152b** of the capacitor **144b** includes a bending end at the bottom thereof, but not limited thereto. In another embodiment, the bottommost surface of the bottom electrode layer **152c** of one capacitor **144c** is lower than the bottommost surfaces of the bottom electrode layers **152a**, **152b** of the capacitors **144a**, **144b**, with the bottom electrode layer **152a** of the capacitor **144a** and/or the bottom electrode layer **152b** of the capacitor **144b** includes a bending end at the bottom thereof. Through these arrangements, the difference of the elemental density between the cell region **104** and the peripheral region **102** will also be balanced while fabricating the semiconductor device **10**, so as to

improve the possible structural defects of the semiconductor device caused by continuously increased cell-density. Then, the semiconductor device **10** of the present embodiment enables to obtain a more reliable structure, to achieve better functions and performances.

[0024] Precisely speaking, further in view of FIG. 2, the capacitor structure **150** includes the bottom electrode layer **152**, the capacitor dielectric layer **154**, and the top electrode layer **156** stacked in sequence in the vertical direction D3. In the present embodiment, the bottom electrode layer **152** for example includes a continuously extended U-shaped cross-section, the capacitor dielectric layer **154** is disposed on the sidewalls and the bottom surface of the U-shaped cross-section of the bottom electrode layer **152**, and the top electrode layer **156** is filled in the gap of the U-shaped cross-section of the bottom electrode layer **152**. Accordingly, each capacitor **144** disposed within the peripheral region **102** will substantially include a cylinder cross-sectional structure, but not limited thereto. It is noted that, the bending ends of the bottom electrode layers **152a**, **152b** of the capacitors **144a**, **144b** are for example disposed between the top surface of the substrate **100** and the first supporting layer **162**, or between the first supporting layer **162** and the second supporting layer **164**, with each bending end gradually bent toward another capacitor **144c** adjacent thereto. In one embodiment, the bending ends of the bottom electrode layers **152b**, **152a**, **152c** are preferably not in physical contact with the bottom electrode layers **152b**, **152a**, **152c** of the capacitors **144b**, **144a**, **144c** being adjacent thereto, and the capacitor dielectric layer **154** covering on the aforementioned bending ends is in physical contact with the capacitor dielectric **154** covering on the bottom electrode layers **152b**, **152a**, **152c** of the capacitors **144b**, **144a**, **144c** being adjacent thereto, as shown in FIG. 2, but not limited thereto.

[0025] The semiconductor device **10** further includes a plurality of buried gate structure **120** disposed within the substrate, and a plurality of plugs **130** disposed on the substrate **100**, both within the cell region **104**. Precisely speaking, the gate structures **120** are extended along a direction (not shown in the drawings) in the substrate, to intersect a shallow trench isolation **112** and a plurality of active areas **114** in the substrate **100**. Each of the gate structures **120** further includes a dielectric layer **122**, a gate dielectric layer **124**, a gate layer **126**, and a capping layer **128** stacked in sequence, with a surface of the capping layer **128** being coplanar with a top surface of the substrate **100**, so that, the gate structures **120** may therefore function like buried word lines (BWLs) of the semiconductor device **10**. Each of the plugs **130** further includes a storage node contact **132** and a storage node pad **134** disposed on the substrate **100**, and two adjacent ones of the plugs **130** are isolated from each other through a storage node contact isolation (SCISO) **136** right above each gate structure **120**. In one embodiment, the storage node contact **132** for example includes an epitaxial material like silicon, silicon phosphate (SiP), silicon germanium (SiGe), or germanium, the storage node pad **134** for example includes a low-resistance metal material like aluminum (Al), titanium (Ti), copper (Cu), or tungsten (W), but not limited thereto. In another embodiment, the storage node pad **134** may optionally include the same conductive material as that of the storage node contact **132**, but is not limited thereto.

[0026] Through these arrangements, each capacitor **142** disposed within the cell region **104** enables to electrically connect to the transistor element within the substrate **100** through the plug **130** disposed underneath, and the capacitors **142** and the plugs **130** will together serve as a smallest memory cell of a storage cell array in a dynamic random access memory (DRAM, not shown in the drawings), for receiving voltage signals from bit line (BL, not shown in the drawings) and the word lines (namely, the gate structures **120**). On the other hand, each capacitor **144** disposed within the peripheral region **102** only contacts the insulating layer **138** underneath, and which cannot electrically connect to the transistor element within the substrate **100**, thereby serving as a dummy capacitor. It is noted that although the bit lines are not illustrated in the drawings of the present disclosure, those who skilled in the arts should fully realize the bit lines are parallel with each other in a direction being perpendicular to the word lines (namely, the gate structures **120**), with a bit line contact (BLC) disposed underneath to electrically connect to the substrate **100**, and with an

insulating layer (for example including an oxide-nitride-oxide structure, not shown in the drawings) disposed on the substrate **100** to electrically isolating the gate structures **120** within the substrate. [0027] According to the semiconductor device **10** of the first embodiment, the capacitors within the peripheral region **102** are physically in contact with each other through the connecting bottom electrodes **152**, the connecting capacitor dielectric layers **154**, and/or the connecting top electrode layers **156**, so as to configure as the dummy capacitors thereby. Then, the difference of the elemental density between the cell region **104** and the peripheral region **102** will be balanced through arranging the dummy capacitors, so as to improve the possible structural defects of the semiconductor device **10** caused by continuously increased cell-density. In this way, the semiconductor device **10** is allowable to obtain a more reliable structure, to achieve better functions and performances.

[0028] Those of ordinary skill in the art should easily realize the semiconductor device in the present disclosure is not limited to the aforementioned embodiment, and which may include other examples or varieties. The following description will detail the different embodiments of the semiconductor device in the present disclosure. To simplify the description, the following description will detail the dissimilarities among the different embodiments and the identical features will not be redundantly described. In order to compare the differences between the embodiments easily, the identical components in each of the following embodiments are marked with identical symbols.

[0029] Please refer to FIG. 3 and FIG. 4, which are schematic diagrams illustrating a semiconductor device **20** according to the second embodiment of the present disclosure, with FIG. 3 and FIG. 4 illustrating a top view and a cross-sectional view of the semiconductor device **20** respectively. The difference between the semiconductor device **20** of the present embodiment and the semiconductor device **10** of the aforementioned embodiment is mainly in that the capacitor structure **250** includes a bottom electrode layer **252**, a capacitor dielectric layer **254**, and a top electrode layer **256** stacked in sequence in the horizontal direction D1, to form a plurality of capacitors **240**, with each capacitor **240** having a cylinder cross-section, but not limited thereto. The bottom electrode layer **252** for example includes a columnar bottom electrode, and the capacitor dielectric layer **254** and the top electrode layer **256** cover the outer side of the bottom electrode layer **252** in sequence, surrounding the columnar bottom electrode thereby.

[0030] The capacitors **240** are arranged in the horizontal direction D1 and the vertical direction D2 into an array, with the supporting structure **160** being disposed between the capacitors **242** within the cell region **104** for isolating and supporting thereto. Each of the capacitors **242** may be electrically connected to the transistor element within the substrate **100** through the plug **130** underneath (as shown in FIG. 4), to form the storage cell array. It is noted that, at least two capacitors **244**, **246** disposed within the peripheral region **102** physically contact with each other in the horizontal direction D1, to function like a dummy capacitor. In one embodiment, two capacitors **244** within the peripheral region **102** are physically in contact with each other through the connecting bottom electrodes **252**, the connecting capacitor dielectric layers **254**, and/or the connecting top electrode layers **256**, to configured as the dummy capacitors as shown in FIG. 3, but not limited thereto. Alternately, at least one capacitor **246** within the peripheral region **102** includes a relative smaller width W1 in the horizontal direction D1, and which enables to directly contact other capacitors **244** adjacent thereto. Through these arrangements, two or more than two capacitors **244**, **246** within the peripheral region **102** contact with each other through the connecting top electrode layers **256**, the connecting capacitor dielectric layer **254**, or the connecting bottom electrode layers **252**, to form the dummy capacitors.

[0031] On the other hand, as shown in FIG. 4, the bottommost surfaces of the bottom electrode layers of at least two capacitors **244a**, **244b**, **244c** to a top surface of the substrate **100** are in different heights h1, h2, h3 in the vertical direction D3. Precisely, the bottommost surface of the bottom electrode layer **252a** of the capacitor **244a** is lower than the bottommost surface of the

bottom electrode layer **252b** of another capacitor **244b**, with the bottom electrode layer **252a** of the capacitor **244a** and/or the bottom electrode layer **252b** of the capacitor **244b** including a bending end at the bottom thereof, but not limited thereto. Alternately, the bottommost surface of the bottom electrode layer **252c** of the capacitor **244c** is lower than the bottommost surface of the bottom electrode layer **252a**, **252b** of the capacitors **244a**, **244c**, with the bottom electrode layer **252a** of the capacitor **244a** and/or the bottom electrode layer **252b** of the capacitor **244b** including a bending end at the bottom thereof.

[0032] According to the semiconductor device **20** of the present embodiment, the capacitors **242** within the peripheral region **102** still physically contact with each other through the connecting bottom electrode layers **252**, the connecting capacitor dielectric layer **254**, and/or the connecting top electrode layers **256**, to serve as the dummy capacitors. With these arrangements, the difference of the elemental density between the cell region **104** and the peripheral region **102** will be balanced through arranging the dummy capacitors, so as to improve the possible structural defects of the semiconductor device **20** caused by continuously increased cell-density. In this way, the semiconductor device **20** is allowable to obtain a more reliable structure, to achieve better functions and performances.

[0033] Please refer to FIG. 5, which is a schematic diagram illustrating a semiconductor device **30** according to the third embodiment of the present disclosure. The structure of the semiconductor device **30** is substantially the same as that of the semiconductor device **10** of the aforementioned first embodiment, with all the similarities being not redundantly described hereinafter. The difference between the semiconductor device **30** of the present embodiment and the semiconductor device **10** of the aforementioned first embodiment is mainly in that the capacitor structure **350** includes a bottom electrode layer **352**, a capacitor dielectric layer **354**, and a top electrode layer **156** stacked in sequence, to form a plurality of capacitors **142**, **344** within the cell region **104** and the peripheral region **102**. It is noted that the bottom electrode layer **352a**, **352b** of at least one of the capacitors **344a**, **344b** within the peripheral region **102** includes a discontinuously film at the bottom thereof.

[0034] Precisely speaking, the bottom electrode layer **352a** of the capacitor **344a** and the bottom electrode layer **352b** of the capacitor **344b** each includes two portions that gradually bending toward an adjacent capacitor **144c**, **344b** at the same side, such that, the bottoms of the two portions are not connected with each other. Then, the bottom electrode layers **352a** and **352b** of the capacitors **344a** and **344b** are respectively in a discontinuous structure, as shown in FIG. 5. Furthermore, the subsequent capacitor dielectric layer **354** covering on the two portions are also not connected with each other, to present in a discontinuous structure as shown in FIG. 5, but not limited thereto. However, in another embodiment, the capacitor dielectric layer **354** covering on the two portions may physically contact the capacitor dielectric layer **354** covering on the bottom electrode layer **352**, **352a**, **152** of an adjacent capacitor **344**, optionally. According to the semiconductor device **30** of the present embodiment, the capacitors **342** within the peripheral region **102** still physically contact with each other through the connecting bottom electrode layers **352**, the connecting capacitor dielectric layer **354**, and/or the connecting top electrode layers **156**, to serve as the dummy capacitors. With these arrangements, the difference of the elemental density between the cell region **104** and the peripheral region **102** will be balanced through arranging the dummy capacitors, so as to improve the possible structural defects of the semiconductor device **30** caused by continuously increased cell-density. In this way, the semiconductor device **30** of the present embodiment is allowable to obtain a more reliable structure, to achieve better functions and performances.

[0035] Please refer to FIG. 6, which is a schematic diagram illustrating a semiconductor device **40** according to the fourth embodiment of the present disclosure. The structure of the semiconductor device **40** is substantially the same as that of the semiconductor device **10** of the aforementioned first embodiment, with all the similarities therebetween being not redundantly described

hereinafter. The difference between the semiconductor device **40** of the present embodiment and the semiconductor device **10** of the aforementioned first embodiment is mainly in that the capacitor structure **450** includes a bottom electrode layer **452**, a capacitor dielectric layer **454**, and a top electrode layer **156** stacked in sequence, to form a plurality of capacitors **142**, **444** within the cell region **104** and the peripheral region **102**. It is noted that the bottom electrode layer **452a** of at least one capacitor **444a** within the peripheral region **102** includes a bending end at the bottom thereof, being gradually bent toward another capacitor **144b**.

[0036] Precisely speaking, the bottom electrode layer **452s** of the capacitor **444a** includes the bending end that gradually bends toward the adjacent capacitor **144b** at one side, while the bottom electrode layer **152b** of the capacitor **144b** also includes a bending end that gradually bends toward the capacitor **444a**. That is, the bending ends of the bottom electrode layer **452a** of the capacitor **444a** and the bottom electrode layer **152b** of the capacitor **144b** are opposite to each other and gradually approach to each other. The bending end of the bottom electrode layer **452a** of the capacitor **444a** is located between the surface of the substrate **100** and the first support layer **162**, while the bending end of the bottom electrode layer **152b** of the capacitor **144b** is located between the first support layer **162** and the second support layer **164**, as shown in FIG. **6**. Also, since the bending ends of the bottom electrode layers **152b** of the capacitor **444a** and the capacitor **144b** gradually approach each other, the capacitor dielectric layers **454** covering on the bottom electrode layer **452a** of the capacitor **444a** and the bottom electrode layer **152b** of the capacitor **144b** are in directly contact with each thereto, but not limited thereto. However, in another embodiment, the bottom electrode layers of the capacitor **444a** and the capacitor **144b** may also be in directly contact (not shown in the drawing), optionally. According to the semiconductor device **40** of the present embodiment, the capacitors **442** within the peripheral region **102** still physically contact with each other through the connecting bottom electrode layers **452**, the connecting capacitor dielectric layer **454**, and/or the connecting top electrode layers **156**, to serve as the dummy capacitors. With these arrangements, the difference of the elemental density between the cell region **104** and the peripheral region **102** will be balanced through arranging the dummy capacitors, so as to improve the possible structural defects of the semiconductor device **40** caused by continuously increased cell-density. In this way, the semiconductor device **40** is allowable to obtain a more reliable structure, to achieve better functions and performances.

[0037] Please refer to FIG. **7**, which is a schematic diagram illustrating a semiconductor device **50** according to the fifth embodiment of the present disclosure. The structure of the semiconductor device **50** is substantially the same as that of the semiconductor device **10** of the aforementioned first embodiment, with all the similarities being not redundantly described hereinafter. The difference between the semiconductor device **50** of the present embodiment and the semiconductor device **10** of the aforementioned first embodiment is mainly in that the capacitor structure **550** includes a bottom electrode layer **552**, a capacitor dielectric layer **554**, and a top electrode layer **156** stacked in sequence, to form a plurality of capacitors **142**, **544** within the cell region **104** and the peripheral region **102**. It is noted that, the bottom electrode layer **152a** of at least one capacitor **144a** within the peripheral region **102** includes a bending end at the bottom thereof, being gradually bent toward another capacitor **544a** at the bottom thereof.

[0038] Precisely speaking, the capacitor **544a** extends in the vertical direction D3 and is disposed on the substrate **100**, with the bottom electrode layer **522a** includes a continuously extending U-shaped cross-section. The bottom electrode layer **152a** of the capacitor **144a** and the bottom electrode layer **552b** of the capacitor **544b** each includes a bending end gradually bent toward the capacitor **544a**, so that, the vertically extended capacitor **544a** is sandwiched between the capacitor **144a** and the capacitor **544b** with the bending end. In addition, since the bottom electrode layer **552a** of the capacitor **544a** has a relatively small width W2 than that of other capacitors **544**, the bending ends of the bottom electrode layer **152a** of the capacitor **144a** and the bottom electrode layer **552b** of the capacitor **544b** gradually approach and physically contact the bottom electrode

layer **552a** of the capacitor **544a**, with the bottom electrode layer **552a** of the capacitor **544a** being sandwiched therebetween, as shown in FIG. 7. However, in another embodiment, the bending ends of the bottom electrode layer **152a** of the capacitor **144a** and the bottom electrode layer **552b** of the capacitor **544b** may also gradually approach to and not in physically contact with the bottom electrode layer **552a** of the capacitor **544a**, so that, the capacitor dielectric layer covering on the bottom electrode layer **152a** of the capacitor **144a**, the bottom electrode layer **552b** of the capacitor **444b**, and the bottom electrode layer **552a** of the capacitor **444a** will be connected and contacted with each other, but not limited thereto. According to semiconductor device **50** of the present embodiment, the capacitors **542** within the peripheral region **102** still physically contact with each other through the connecting bottom electrode layers **552**, the connecting capacitor dielectric layer **554**, and/or the connecting top electrode layers **156**, to serve as the dummy capacitors. With these arrangements, the difference of the elemental density between the cell region **104** and the peripheral region **102** will be balanced through arranging the dummy capacitors, so as to improve the possible structural defects of the semiconductor device **50** caused by continuously increased cell-density. In this way, the semiconductor device **50** is allowable to obtain a more reliable structure, to achieve better functions and performances.

[0039] Please refer to FIG. 8, which is a schematic diagram illustrating a semiconductor device **60** according to the sixth embodiment of the present disclosure. The structure of the semiconductor device **60** is substantially the same as that of the semiconductor device **10** of the aforementioned first embodiment, with all the similarities therebetween being not redundantly described hereinafter. The difference between the semiconductor device **60** of the present embodiment and the semiconductor device **10** of the aforementioned first embodiment is mainly in that the capacitor structure **650** includes a bottom electrode layer **652**, a capacitor dielectric layer **654**, and a top electrode layer **156** stacked in sequence, to form a plurality of capacitors **642**, **644** within the cell region **104** and the peripheral region **102**. It is noted that the bottom electrode layer **652a** of at least one capacitor **644a** within the peripheral area **102** includes a bending end that gradually bending towards another capacitor **644b**, while the bottom electrode layer **652c** of at least one capacitor **644c** within the cell region **104** also includes a bending end that gradually bends towards the capacitor **644b**, with the bottoms of the bottom electrode layers **652a**, **652c** still in physical contact with the insulating layer **138** or the plug **130** underneath.

[0040] Precisely speaking, the capacitors **642**, **644a** respectively arranged in the cell region **104** and the peripheral region **102** extend in the vertical direction D3, over the substrate **100**. The bending ends of the bottom electrode layers **652c** and **652a** of the capacitors **642** and **644a** are located between the surface of the substrate **100** and the first support layer **162**, to substantially present in a continuously extending U-shaped cross section, as shown in FIG. 8. The bending end of the bottom electrode layer **652a** of the capacitor **644a** is preferably approach to and not in physically contact with the bottom electrode layer **652b** of the adjacent capacitor **644b**, so that, the capacitor dielectric layers **654** covering the bottom electrode layers **652a**, **652b** of the capacitors **644a** and **644b** are connected and contacted with each other, but not limited thereto. However, in another embodiment, the bottom electrode layers **652a**, **652b** of the capacitors **644a**, **644b** may also be in direct contact with each other. Then, the capacitor **642** arranged in the cell region **104** will still be aligned with a corresponding plug underneath, to form a storage cell array, while the capacitors **644a**, **644b** and **144b** arranged in the peripheral region **102** contact the insulating layer **138** to serve as the dummy capacitors.

[0041] According to the semiconductor device **60** of the present embodiment, the capacitors **642** within the peripheral region **102** still physically contact with each other through the connecting bottom electrode layers **652**, the connecting capacitor dielectric layer **654**, and/or the connecting top electrode layers **156**, to serve as the dummy capacitors. With these arrangements, the difference of the elemental density between the cell region **104** and the peripheral region **102** will be balanced through arranging the dummy capacitors, so as to improve the possible structural defects of the

semiconductor device **60** caused by continuously increased cell-density. In this way, the semiconductor device **60** is allowable to obtain a more reliable structure, to achieve better functions and performances. In addition, those skilled in art should fully realize that although the semiconductor device **60** of the present embodiment includes the capacitors **644a**, **644b**, **144b** within the peripheral region **102** serving as the dummy capacitors, the practical structure of the dummy capacitors is not limited to which is shown in FIG. **8**, and which may also include the capacitors **544a**, **544b** as shown in FIG. **7** to serve as the dummy capacitors, the capacitor **444a** as shown in FIG. **6** to serve as the dummy capacitors, or the capacitor **344a**, **344**, **344c** as shown in FIG. **5** to serve as the dummy capacitors, due to the specific product requirements. The bottom electrode layers of the aforementioned capacitors may optionally contact or not contact the bottom electrode layer **652c** of an adjacent capacitor **642**, to obtain a more reliable structure thereby.

[0042] Overall speaking, according to the semiconductor device of the present disclosure, dummy capacitors are arranged around the storage cell array, whose top electrode layers are in physical contact with each other, capacitor dielectric layer are in physical contact with each other, and/or top electric layers are in physical contact with each other, optionally to improve the possible structural defects of the semiconductor device caused by continuously increased cell-density. Then, the semiconductor device of the present disclosure is allowable to obtain a more reliable structure, to achieve better functions and performances.

[0043] Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A semiconductor device, comprising: a substrate; and a capacitor structure, disposed on the substrate and comprising a plurality of capacitors arranged in an array, wherein at least two of the capacitors physically contact with each other in a horizontal direction being horizontal to the substrate.
2. The semiconductor device according to claim 1, wherein each of the capacitors comprises a bottom electrode layer, a capacitor dielectric layer and a top electrode layer disposed in sequence along the horizontal direction.
3. The semiconductor device according to claim 2, wherein the bottom electrode layers of at least two of the capacitors physically contact with each other.
4. The semiconductor device according to claim 2, wherein the top electrode layers of at least two of the capacitors physically contact with each other.
5. The semiconductor device according to claim 2, wherein the capacitor dielectric layers of at least two of the capacitors physically contact with each other.
6. The semiconductor device according to claim 1, wherein at least two of the capacitors are in different widths in the horizontal direction.
7. A semiconductor device, comprising: a substrate; and a capacitor structure, disposed on the substrate and comprising at least one first capacitor and at least one second capacitor, a bottommost surface of a bottom electrode layer of the at least one first capacitor is lower than a bottommost surface of a bottom electrode layer of the at least one second capacitor, and at least one of the at least one first capacitor and the at least one second capacitor comprises a bending end; and a supporting structure, disposed on the substrate and between the at least one first capacitor and the at least one second capacitor, the supporting structure comprising a first supporting layer and a second supporting layer stacked in sequence from bottom to top.
8. The semiconductor device according to claim 7, wherein the bottom electrode layer of the at least one first capacitor bends toward one side of the at least one second capacitor.

9. The semiconductor device according to claim 7, wherein the bottom electrode layer of the at least one second capacitor bends toward one side of the at least one first capacitor.
10. The semiconductor device according to claim 7, wherein a bottommost surface of the bending end of the bottom electrode layer of the at least one second capacitor is disposed between the first supporting layer and the second supporting layer.
11. The semiconductor device according to claim 7, wherein a bottommost surface of the bending end of the bottom electrode layer of the at least one second capacitor is disposed between the first supporting layer and the substrate.
12. The semiconductor device according to claim 7, wherein the bottom electrode layer of the at least one second capacitor comprises a discontinuous bottom structure.
13. The semiconductor device according to claim 7, wherein the at least one first capacitor and the at least one second capacitor comprise a cylinder cross-sectional structure respectively.
14. The semiconductor device according to claim 7, wherein the bending end of the bottom electrode layer of the at least one second capacitor does not physically contact the bottom electrode layer of the at least one first capacitor.
15. The semiconductor device according to claim 14, wherein a capacitor dielectric layer disposed on the bottom electrode layer of the at least one first capacitor physically contacts a capacitor dielectric layer disposed on the bottom electrode layer of the at least one second capacitor.
16. The semiconductor device according to claim 14, wherein a capacitor dielectric layer disposed on the bottom electrode layer of the at least one first capacitor does not physically contact a capacitor dielectric layer disposed on the bottom electrode layer of the at least one second capacitor.
17. The semiconductor device according to claim 7, wherein the bottom electrode layer of the at least one first capacitor physically contacts the bottom electrode layer of the at least one second capacitor.
18. The semiconductor device according to claim 7, wherein a width of the bottom electrode layer of the at least one first capacitor is different from a width of the bottom electrode layer of the at least one second capacitor.
19. The semiconductor device according to claim 7, wherein the bottom electrode layer of the at least one first capacitor comprises the bending end, and the bottom electrode layer of the at least one second capacitor is vertically disposed on the substrate.
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